

1.2. Features

- Bluetooth 5, IEEE 802.15.4, 2.4 GHz transceiver
 - -95dBm sensitivity in 1Mbps Bluetooth low energy (BLE) mode
 - -103dBm sensitivity in 125Kbps BLE mode (long range)
 - +8 dBm TX power (down to -20 dBm in 4 dB steps)
 - On-air compatible with nRF52, nRF51, nRF24L and nRF24AP Series
 - Programmable output power from +8dBm to -20dB
 - RSSI (1dB resolution)
 - Supported data rates:
 - Bluetooth 5: 2 Mbps, 1 Mbps, 500 kbps, 125 kbps
 - IEEE 802.15.4-2006: 250 kbps
 - Proprietary 2.4 GHz: 2 Mbps, 1 Mbps
- ARM Cortex –M4 32-bit processor with FPU, 64 MHz
- Memory: 1MB flash / 256KB RAM
- HW accelerated security
 - ARM TrustZone Cryptocell 310 security subsystem
 - 128 bit AES / ECB / CCM / AAR co-processor (on-the-fly packet encryption)
- Advanced on-chip interfaces
 - USB 2.0 full speed (12Mbps) controller
 - QSPI 32MHz interface
 - High speed 32MHz SPI
 - Type 2 near field communication (NFC-A) tag with wake-on field
 - Programmable peripheral interconnect (PPI)
 - 48 general purpose I/O pins
 - EasyDMA automated data transfer between memory and peripherals.
- 12 bit, 200ksps ADC – 8 configurable channels with programmable gain
- 4 x 4 channel pulse width modulator (PWM) units with EasyDMA
- Audio peripherals : I2S, digital microphone interface (PDM)
- 5 X 32-bit timers with counter mode
- Up to 4 x SPI masters / 3 x SPI slaves with EasyDMA
- Up to 2 x I2C compatible 2-wire masters / slaves
- 2 x UART(CTS/RTS) with EasyDMA
- Quadrature decoder (QDEC)
- 3 x 24-bit real-time counters (RTC)